

SNEHA SONKUSARE

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SUMMARY

Data Scientist with 2+ years of experience building end-to-end ML pipelines, from feature engineering and model training to deployment and monitoring, across financial services and research domains. Proficient in Python, SQL, deep learning (PyTorch, TensorFlow), NLP, and cloud platforms (AWS, GCP). M.S. in Data Science with a track record of building predictive models and feature engineering pipelines that drive measurable business impact.

SKILLS

- **Languages:** Python (Pandas, NumPy, Scikit-learn), R, SQL
- **Machine Learning:** Classification, Regression, Clustering, Random Forest, Feature Engineering, XGBoost, LightGBM, SVMs
- **Deep Learning:** TensorFlow, Keras, PyTorch, CNNs, LSTMs, NLP models
- **Visualization:** Tableau, Power BI, Seaborn, Matplotlib
- **Databases:** MySQL, Snowflake, Amazon Redshift, PostgreSQL
- **Cloud & Tools:** AWS (S3, Glue, SageMaker, QuickSight), Apache Spark, Airflow, dbt, Docker, CI/CD, Streamlit
- **Statistics:** Hypothesis Testing, A/B Testing, Regression Analysis, Bayesian Inference, Price Elasticity Modeling

EXPERIENCE

Graduate Research Assistant | UMass Dartmouth School for Marine Science & Technology New Bedford, MA | Jan 2025 – Dec 2025

- Performed EDA on 500K+ oceanographic records, engineered domain-specific features, and trained Random Forest and CNN models (PyTorch/TensorFlow) achieving 85% accuracy; tracked all experiments via MLflow for full reproducibility.
- Optimized model performance through cross-validation, precision, recall, and F1 tuning; translated findings into actionable insights for non-technical research teams, directly informing conservation decisions.
- Spearheaded standardization of metadata and data governance practices across 5+ research systems that lacked consistent structure cutting data inconsistencies by 25% and ensuring model reproducibility across research and IT teams.
- Developed Apache Spark pipelines for end-to-end sensor data processing, collaborating with marine scientists to align model outputs with research needs and improving data accuracy by 20%.

Data Analyst | Hexaware Technologies LTD Mumbai, India | Sep 2021 – Jan 2024

- Collaborated with finance, operations, and product stakeholders to translate business pain points into data questions, writing complex SQL queries (joins, aggregations, window functions) across 5+ source systems to build a unified data layer, reducing data errors by 30% and saving 10+ hours of weekly manual corrections.
- Detected recurring anomalies in fraud and credit risk data through EDA and statistical hypothesis testing; applied Scikit-learn classification models that improved decision accuracy by 25%, and maintained 8+ Power BI dashboards for fraud detection, risk, and portfolio monitoring, cutting report generation time by 35%.
- Established 15+ KPIs across finance, product, and operations in collaboration with senior stakeholders, delivering self-serve analytics that reduced ad-hoc data requests by 20% and eliminated recurring manual reporting rebuilds for analysts.
- Diagnosed and resolved 20+ pipeline failures in Agile cross-functional teams alongside senior engineers, applying targeted fixes that boosted processing speed by 30% and reduced query runtime by 40%, while clearly communicating issue impact to non-technical stakeholders throughout.

PROJECTS

AI-Powered Learning Platform | [LearnAI](#) Feb 2026 – Mar 2026

- Developed and deployed an AI learning platform using LLaMA 3.3 70B and Groq API, generating personalized courses from a single input with 10+ tools including Mock Interview, Quiz Generator, and AI Doubt Solver.
- Engineered a structured prompt system that consolidated 7 API calls into 1, reducing generation time by 80%, and implemented a 3-model fallback chain for rate-limit resilience with session-state caching across 12-chapter multilingual lessons.

Job Market Intelligence and Career Advisor | [JobMarketIntel](#) Oct 2025 – Dec 2025

- Engineered a real-time pipeline that scrapes hundreds of daily job postings, extracts in-demand skills using NLP, and predicts salaries using XGBoost with SHAP explainability.
- Built a Personal Career Advisor that inputs a user's skillset, predicts their market rate, and recommends the top 3 skills to maximize salary growth powered by a custom Job Market Health Index (0–100 scale).

Instruction Fine-Tuning Based Medical Chatbot | [GitHub](#) Apr 2025 – May 2025

- Fine-tuned DeepSeek-R1 (LoRA/QLoRA) on medical Q&A and built a Neo4j-backed RAG pipeline for structured clinical data retrieval, improving response accuracy by 30%.
- Developed a production-ready inference pipeline using Flask REST API with real-time model monitoring for latency and drift, containerized and configured for AWS deployment.

EDUCATION

UNIVERSITY OF MASSACHUSETTS DARTMOUTH Jan 2024 – Dec 2025

Master of Science in Data Science | GPA : 3.73

(Relevant Coursework: Artificial Intelligence, Advanced Machine Learning, Big Data Analytics, Database Design).

MAHARASHTRA INSTITUTE OF TECHNOLOGY

Aug 2017 – Jul 2021

Bachelor of Technology – Electronics and Communication